The Missing Amplitude

An audit of the discrete universe program: what one reversible rule on the hyperbolic 24-cell mesh produces, what it cannot, and where to look

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Abstract

The discrete universe program models reality as one growing hyperbolic honeycomb, $\{3,4,3,4\}$, whose cells are 24-cells, carrying a ternary value updated by one reversible, charge-conserving local rule. This paper reports an adversarial audit of the 832 experiments behind that program. Under a grading in which a result is emergent only when the base rule produced it against a computed control, 45 qualify, down from 92. The central finding is exact: the rule is a permutation of ternary slots with no amplitude. Its vacuum flashes with period three, a seeded defect neither spreads nor superposes, its lift to superpositions of configurations is a permutation matrix with zero cross term, and so is its coarse map on two-tone pairs. The one amplitude it does supply, a $\mathbb{Z}_3$ sum over the mesh, cancels on the vacuum and locks every defect to the same clock phase, so nothing interferes. Every quantum result in the program runs on a hand-written walk or a small Hilbert space and is graded so. New exact results on the mesh include the lattice Ward identity with its three Wilson lines, a toric code with four logical qubits per component, the discovery that an even-sided periodic mesh is two lattices (which had produced two false passes), and the Ehrenfest theorem on the walk to 10^{-13} of a cell. The corrections to the earlier papers are listed, and the paper closes with the open problems, the constraints that make them finite, and the directions worth taking. Claude Code was used throughout, with Claude Opus 4 and Opus 5 on the earlier experiments and Claude Fable 5 on this audit, to write the code and this paper.

The Claim Under Audit

The program's claim is that five things suffice. A single substance carrying a ternary tone in $\{-1,0,+1\}$. The four-dimensional hyperbolic honeycomb $\{3,4,3,4\}$ as the arrangement of that substance, each cell a 24-cell whose twenty-four neighbours are the D_4 root system. One local law, a collision followed by a stream along each of the twenty-four directions, applied once per beat, exactly reversible and conserving a $U(1)$ charge. Growth at the open edge of the mesh, which supplies the arrow. And nothing else. The earlier papers [8, 9] argued that spinors, one generation of matter with the right charges, the Standard Model gauge group, the Dirac dispersion, Lorentz invariance, gravity as an equation of state, holography, the Born rule and the Tsirelson bound, a deterministic cosmology, and universal computation follow from those five things, each backed by a computational experiment.

The audit asks one question of every experiment: what did the base rule actually do here. An experiment can print a green result while the rule played no part in it. A textbook formula can be evaluated on a substrate that was never built. A walk can be written by hand, given the mesh's coin as its rotation, and called the coin's own dynamics. A control can be a literal `false` typed into the file. A verdict can rest on a mesh that is two disconnected lattices without anyone noticing. Each of these happened in the suite, and each is a way for a program to fool itself while every test passes. The audit found them, fixed them, added the checks that keep them fixed, and then asked what is left.

What is left is a base that is well defined, exactly reversible, and productive. It produces a light cone with a measured speed, hyperbolic geometry with a flat cusp, a genesis attractor on a growing mesh, conserved charges, a cell complex that carries quantum codes, rotational isotropy restored to fourth order by the D_4 symmetry, and an associative memory whose search cost is logarithmic in the store. It does not produce interference, and the audit proves that no phase on its configurations can. That proof, and the one coarse amplitude that escapes it, is the paper's centre. The rest of the paper is the method that found it, the corrections it forces on the earlier papers, and the open problems, stated with the constraints that make them tractable.

The Base

The mesh. The committed substrate is $\{3,4,3,4\}$, the honeycomb of hyperbolic 4-space by 24-cells, with four cells around each triangle. The twenty-four neighbours of a cell are the roots of D_4, the vectors $(\pm1,\pm1,0,0)$ and their permutations. The flat ideal boundary at a cusp is the cubic honeycomb $\{4,3,4\}$ of ordinary space. The experiments build this mesh three ways: the exact reflection-group orbit engine, which grows chambers outward from a seed and reports the cell graph, a cell-direct builder with a cell cap, and a periodic D_4 lattice on a four-torus of side s, which is $\mathbb{Z}^4$ with the twenty-four root adjacencies and is the flat model of the local structure. Comparison substrates ($\{5,3,4\}$, $\{7,3\}$, $\{5,3\}$, cubic lattices, Bethe trees, scrambled graphs of matched degree) appear only as controls.

The tone and the will. Each cell holds one ternary slot per direction, an `Int8` array of $24 \times$ cells entries in $\{-1,0,+1\}$. The state of the whole mesh is this array. Charge is its sum.

The knit. One beat is a collision at every cell followed by a stream, each slot's value moving one cell along its direction. Two collisions are committed. The charge rule acts on each pair of opposite slots through a nine-state table with create, flip and annihilate moves, conserving the pair's charge. The momentum rule rotates a head-on pair of equal signs into a perpendicular pair, conserving charge and the D_4 momentum. Both are bijections on the slot array, so every beat is a permutation of the finite state space, and running the inverse collision after the inverse stream recovers the start exactly. This is the property every experiment on reversibility measures, and it holds to an integer Hamming distance of zero.

The arrow. The mesh grows one shell per beat at its frontier. Fresh cells are born at peace, all slots zero, and creation happens only where the moving frontier meets the older structure. Because the frontier never returns to a cell, growth alone desynchronises the update and supplies the direction of time. No rate parameter and no random number enter the genesis experiments.

A fact about the periodic mesh. The D_4 roots all have even coordinate sum, so on a four-torus of even side the cells of even and odd coordinate sum never meet. A periodic D_4 lattice of side 4, 6 or 8 is two disconnected lattices of 128, 648 and 2048 cells, and a breadth-first search from any cell reaches exactly half of the mesh. An odd side wraps the two classes together and the mesh is connected. The audit found this through a quantum code (Section 6), traced it through the forty even-sided call sites in the suite, and found two passing experiments that depended on it (Section 7).

2.1. The two collisions, exactly

Both committed collisions act on the twelve lines of a cell, a line being a direction paired with its opposite, and both are stated here in full so that the permutation claim of Section 5 can be read off.

The charge rule applies one table to the two tones (a, b) meeting head-on on each line:

pair	becomes	move
$(-1,-1)$, $(+1,+1)$	unchanged	like signs are inert
$(-1,0)$, $(0,-1)$, $(+1,0)$, $(0,+1)$	swapped	a charge hops past a peace
$(0,0)$	$(+1,-1)$	peace creates a balanced pair
$(+1,-1)$	$(-1,+1)$	the pair flips
$(-1,+1)$	$(0,0)$	annihilation closes the cycle

The table is a bijection on the nine pair states that conserves $a + b$. The create-flip-annihilate moves form a 3-cycle on the three charge-zero states, and that 3-cycle is the period-three flash of the vacuum: an all-zero mesh becomes all $(+1,-1)$, then all $(-1,+1)$, then all zero. The rule is not an involution, so its inverse is the inverted table, which the engine runs after the inverse stream to go backward in time.

The momentum rule pairs the twelve lines into six and rotates a head-on pair of equal signs into an empty partner line: (s, s) on line i with $(0, 0)$ on line j becomes $(0, 0)$ on i and (s, s) on j, and the reverse, for $s = \pm 1$. Everything else is fixed. It is its own inverse, conserves charge, and conserves the D_4 momentum because the two lines of a pair are perpendicular roots. Its vacuum is a fixed point, and a lone tone never meets the condition, which is why it streams ballistically forever.

Neither table contains a number outside $\{-1, 0, +1\}$. Composed with the stream, each is a permutation of the 3^{24N} states of an N-cell mesh. That is the whole of the base dynamics.

The Method

Every experiment declares a depth, and the audit made the four grades mean something exact.

- **L0, circular.** The answer was put in by hand. A typed constant that reaches the verdict, an integrated equation whose coefficient is then reported as a discovery. Proves nothing, kept only as a consistency note.

- **L1, known mathematics.** An established fact correctly confirmed. The twenty-four directions form the binary tetrahedral group. A textbook formula evaluated with no mesh built.

- **L2, known physics on a stated model.** A known construction run on this system and reproducing what it is known to produce. A coined Dirac walk, a lattice gauge theory, a tight-binding chain, a small Hilbert space.
- **L3, emergent.** One base rule produced the result as a measured consequence on a substrate the experiment built, against a control that was computed and could have said no.

Two further fields were tightened. The substrate label names a mesh the experiment builds or runs on, or the coin algebra of that mesh, and an experiment with none of that in its transitive import graph or its own code carries no substrate label. The control field holds numbers the experiment computed. A literal or a typed formula is not a control. Both are now enforced by checks that run with the suite, and a third check keeps the coverage map in step with the catalog.

What the checks found. Of 92 experiments graded L3, 40 touched no substrate, no rule and no coin algebra anywhere in their import graph. Of 817 experiments, 166 declared the $\{3,4,3,4\}$ substrate with nothing related to it imported, and 129 of those were relabelled after fourteen were read by hand, with 20 octonion-algebra files held for review. Twenty-four literal constants in twenty files reached a verdict, among them a control for proton decay that was the word `false`. Four registered experiments had never been imported by the suite's barrel. The type checker, which the runner never invoked, reported eleven errors. Every one of these is now zero, and the counts stand at 832 experiments in 825 files: 14 at L0, 209 at L1, 564 at L2, 45 at L3, of which 39 are marked for publication.

Robustness. A verdict that holds at one size is a knife edge until shown otherwise. The harness now carries a scale, and 44 of the 45 emergent results run at half, default and one and a half times their size, 27 through the scale and 17 through their own size parameters. The one exception evaluates a dispersion on the twenty-four roots and has no size. Four flips came out. One was a measurement artifact, a frequency read off in whole beats that was exact only because every default wavelength had an integer half-period, corrected with an interpolated transform. Two are size floors, stated in the experiments. One is a genuine knife edge, an adaptive-fill experiment whose largest self-organised patch grows from ten cells to fifteen against a strict threshold of fifteen. It fails at its default size and passes at one and a half times, and its threshold was not moved. The scale check also caught its own blind spot: five experiments built their Coxeter mesh with a depth that bound the size before the chamber cap did, so the first scaled run moved nothing. Those builds were changed to the cap that reproduces the default mesh cell for cell, and rerun with the size actually changing.

Identity under refactoring. The audit removed duplicated code from the library, one coined-walk stepper in place of six copies, one complex-pair module in place of seven, one opposite-direction helper in place of seventy inline copies, and smaller pairs. Each hoist was proved by running every affected experiment before and after and diffing status, metrics and control. The diffs are empty, apart from the one change made on purpose.

Determinism. The base admits no random number. Where an experiment's initial condition is a hashed or seeded fill, its notes say so and call the result ensemble-style, with robustness from the size sweep rather than from seeds. Fourteen of the emergent results carry such a fill, and replacing the fills with structured patterns is open work.

3.1. The audit in numbers

	before	after
experiments in the catalog	820	832
never imported by the suite	4	0
graded L3	92	45
L3 marked for publication	80	39
graded L2	526	564
graded L1	188	209
graded L0	14	14
$\{3, 4, 3, 4\}$ labels with no evidence	166	0, 20 held
typed constants reaching a verdict	24	0
type-checker errors	11	0
emergent results perturbed in size	0	44 of 45
copies of the coined walk	6	1
inline copies of one mesh helper	70	0

3.2. Coverage by arena

Depth by arena after the audit, with the count marked for publication. The zeros in the L3 column are the arenas with no emergent result, and quantum and spin are zero because every former L3 there ran a hand-written walk or a textbook calculation.

arena	L0	L1	L2	L3	paper
selves	3	14	144	14	109
gauge	3	29	34	6	43
foundations	0	37	53	0	56
quantum	1	19	74	0	67
gravity	4	11	37	1	23
cosmology	1	11	32	7	37
spin	1	17	25	0	30
relativity	1	4	35	3	34
geometry	0	13	23	2	20
holography	0	8	24	4	18
substrate survey	0	17	12	1	13
data structure	0	16	12	0	28
associative	0	2	9	5	16
renormalization	0	3	12	2	8
computation	0	2	12	0	10
fluids	0	0	13	0	11
addressing	0	5	8	0	3
method	0	1	5	0	3
total	14	209	564	45	528

3.3. The perturbation results

Every emergent result was rerun at half, default and one and a half times its size. The table lists the ones whose verdict or headline number moved. The others held every verdict boolean at all three sizes.

experiment	what moved
propagating mode	speed squared read as exactly 1 at the default size and failed at 1.5×: a frequency read in whole beats, exact only for integer half-periods. With an interpolated transform, 0.959, 0.980, 0.985 at sides 12, 24, 36
capacity versus curvature	fails at 600 cells, where the control's coverage radius ties, holds at 1200 and 1800
selves at scale	fails at 30000 cells (one patch over fifty), holds at 60000 and 90000
self-emergence	the largest adaptive-fill patch grows 10 to 15 cells against a strict 1.5× threshold. Fails at 709 and 1316 cells, passes at 1792. Threshold not moved
global-workspace ignition	broadcast exactly 0.5 on the split even-sided mesh, sub-threshold seed 0.02. On connected meshes broadcast 1.0, sub-threshold 0.26. Now fails
recordability ceiling	plateau flatness 0.013 on the split mesh, 0.173 at sides 9, 11, 13 and 15. Now fails
anesthetic collapse	passes at side 6, fails at 5 (the core fills the torus) and at 7 (awake margin 0.13). Noted, not tuned

Five of the fourteen experiments converted in the last slice built their mesh with a depth that capped the size before the chamber cap did, so their first scaled run held on an unchanged mesh. The audit's own check had a hole, and the fix (a cap that reproduces the default mesh exactly, verified cell for cell, then scaled) is the reason the table above can be trusted.

What the Rule Produces

These are the results that survive at L3: a substrate built, a dynamics run on it, a computed control, and a verdict that holds across sizes. The numbers are those the experiments print.

Genesis is an attractor of the growing mesh. On a $\{5, 3, 4\}$ mesh of 1316 cells grown by the reflection engine, the charge dynamics reached from the void and from a dense alternating start agree in density (0.322 against 0.330 of cells charged), in net charge (zero, conserved), and in same-sign neighbour fraction (0.488 against 0.460). Four different charge-zero starts, the void, a pair seed, a dense start and a charged block, converge to the same living fraction under the arrow and all collapse to the dead attractor with the arrow off. The same holds at 709 and 1792 cells. On a genuinely growing mesh with no rate parameter, no creation hash and no arrow parameter, the moving frontier alone polarises the fresh peace it meets, a living charge-zero universe emerges and sustains after growth completes, and a static all-peace mesh stays dead.

On the committed $\{3,4,3,4\}$ mesh the same genesis runs on purely ternary integer structure with no decimal and no randomness, at 646, 1033 and 1482 cells.

Only negative curvature sustains the arrow. A spherical reflection group ($\{5,3\}$) has a finite orbit that closes, its frontier vanishes and its wake cannot sustain creation, while the hyperbolic groups ($\{7,3\}$, $\{5,3,4\}$) have orbits that grow geometrically, a perpetual frontier, and life sustained. The permanence of the arrow traces to the sign of the curvature.

A light cone with a measured speed. A tone excitation on the momentum rule propagates ballistically on the periodic D_4 lattice. Its measured speed squared is 0.959, 0.980 and 0.985 at sides 12, 24 and 36, converging on one, and the momentum phase speed spread is under 0.01 while the charge rule's is 1.44 (the control). Wavelengths under six cells are excluded as the lattice-dispersion regime, where the phase speed falls to 0.73. On the curved $\{5,3,4\}$ bulk a second-order integer field propagates with a finite front speed, is exactly reversible (forward forty beats then backward forty recovers the start with zero error), stays bounded, and keeps all three properties with a nonlinear self-coupling on, while a first-order integrator cannot recover.

Rotational isotropy to fourth order, deterministically. The twenty-four D_4 directions satisfy $\sum_d d_i^4 = 3\sum_d d_i^2 d_j^2$, so the wave dispersion $\omega^2(k) = \sum_d (1 - \cos k \cdot d)$ is isotropic through order four and its axis-versus-diagonal anisotropy scales as q^6, measured, against q^2 on a cubic lattice. Both lattices have even dispersion and a vanishing linear Lorentz-violation coefficient, so the gamma-ray bound on that coefficient does not by itself separate them (residuals of order 10^{-109} and 10^{-37} at a Planck-scale cell), and the real difference is the order of isotropy. No random sprinkling enters.

The exact Ward identity on the lattice. The curl-curl operator of the lattice Maxwell field on the three-torus annihilates every pure-gauge gradient field to 9×10^{-16} and has exactly sites $+\,2$ zero modes (66, 127 and 218 at sides 4, 5 and 6): one gauge mode per site, less the constant, plus the three constant link fields that are closed but not exact on the torus, its Wilson lines. A Proca mass term shifts every mode by exactly m^2 and leaves no zero mode. The first draft predicted sites $-\,1$ and the measurement corrected it.

Associative memory scales logarithmically. On the $\{3,4,3,4\}$ bulk the content search costs one pass per word trit, independent of the store, and the communication radius grows by one when the store grows eightfold, while a cubic memory's radius grows polynomially. Attractor recall is not a base capability: a dissipative Hopfield layer cleans a corrupted cue to its prototype while the bare reversible rule, on the same cue, stays near chance because it conserves charge and has no energy to descend. The negative is the result.

Selves on the flat layer. The boundary-to-volume ratio of a ball falls with radius on the flat horosphere and stays near constant in the bulk, so a compact low-leak self is geometrically possible on the flat layer and not in the bulk, and the same emergent self leaks less and holds more of itself unmaintained there. The flat layer is built directly at fifteen million cells, a thousand times the affordable bulk, with charge exactly conserved. A self's causal structure survives two recursive coarse-grainings where a random map loses it, and a cohesive cell is a local maximum of tone integration whose value collapses when the fills across its middle are cut while the wiring stays fixed, so the measure reads the dynamics and not the graph.

What the cell complex carries. The D_4 complex of the periodic mesh, with vertices, root edges and root triangles, supports a toric code: qubits on the 972 edges of the side-3 torus, X checks on cells and Z checks on triangles, every pair commuting, with exactly four logical qubits by exact rank over $\mathbb{F}_2$, and zero on an open patch. At side 4 the count is eight, which is how the two-component fact of the even side was found.

The Missing Amplitude

The program's quantum results all assumed one thing: that the coined Dirac walk on the D_4 coin is the single-particle sector of the base rule. It is not, and the five experiments below settle it exactly.

The rule is a permutation with no amplitude. The state is an `Int8` array, the collision is a table lookup, the stream is a relabelling. There is no complex number anywhere in a beat. The charge rule's vacuum, the all-zero array, is not fixed: the create move fires everywhere and the vacuum flashes with period three, measured on a side-8 lattice. A single seeded tone stays a defect of at most two slots and radius two over the run, and two seeded defects evolve as the exact set-union of their separate runs. The coined walk, given the same seed, spreads to a support of five sites in the same beats with a site-wise cross term of 0.357. Nothing in the rule spreads and nothing superposes.

The momentum rule's lone tone is a classical particle. Under the momentum-conserving collision the vacuum stays empty, a lone tone occupies exactly one slot and moves exactly one cell per beat in a straight line, reaching radius four in four beats. Two tones on parallel or crossing lines are the exact union of their separate runs, and opposite-sign head-on tones pass through each other with two slots and charge zero, no annihilation. This is ballistic motion with no dispersion, no spreading and no interference, at every size.

No phase on configurations can interfere. Suppose one lifts the rule to the space of complex superpositions of configurations, as the Koopman construction does for any deterministic map. The lift of a bijection is a permutation matrix. The experiment pushes 49 configurations, each carrying a phase, through four beats of the rule and finds 49 distinct images, every weight exactly relabelled, cross term zero, norm 49 before and after. The control is a conserving but irreversible sorting collision, which merges the same 49 branches into three images with a cross term of 24 and does not conserve the norm. Interference needs the map to send distinct inputs to the same output with different phases. A permutation never does. This closes the whole class of proposals that would attach a phase to the microstate.

No phase on pairs at a cell can interfere either. The natural coarser object is the pair of tones meeting at a cell. The induced map of the momentum rule on the 1104 two-tone states at one cell is again a permutation: 1104 images, 24 states changed by the head-on rotation, everything else fixed. The sorting control gives three images.

The one amplitude the rule supplies. Take the $\mathbb{Z}_3$ character of the tone, ω^t with $\omega = e^{2\pi i/3}$, summed over every slot of the mesh. On the charge rule this coarse amplitude has exact properties. The flashing vacuum sums to zero over its three-beat period, to 1.8×10^{-12}. A defect keeps magnitude $\sqrt{3}$ (1.732051 at every beat) with a phase locked to the vacuum clock, $30^\circ, 30^\circ, 150^\circ$ with period three. Two defects add exactly (additivity to 2×10^{-16}). And the exact negative: two defects have relative phase zero at every beat, so their cross term is $6 = 2 \times 3$, maximal, and their intensities add with no interference. Every defect is born in the same beat of the same clock. The momentum rule's vacuum shows neither the cancellation nor the clock (magnitude 15000, one phase), the control. This is the sharpest statement of the open problem the program can make: the rule does carry a dynamical phase, it is the vacuum clock, and it is the same phase everywhere.

Everything with a ψ in it therefore runs on a model. The coined Dirac walk is a hand-written unitary map on two complex amplitudes per site, coin then shift, and the program's tunnelling, zitterbewegung, Klein paradox, Bloch oscillation, Aubry-André localisation, Jackiw-Rebbi mode, winding number and bulk-boundary results are correct reproductions of known quantum-walk physics on it [5, 15, 2, 16, 7]. The Born-rule, Bell, CHSH, GHZ and contextuality results are exact statements about small Hilbert spaces and their symmetries. The area law is a free-fermion chain. Each is now graded L1 or L2, carries its prior art, and names the model it runs on. None of them is evidence that the base is quantum.

What Is Reproduced on Models, Exactly

Reproductions are worth having when they are exact and when the model is named. Three added by the audit are of that kind, and each taught the program something about its own objects.

The toric code from the mesh complex. The periodic D_4 lattice has a natural cell complex: cells as vertices, the twenty-four root directions as edges, and the triangles closed by pairs of roots whose sum is a root. Qubits on edges, an X check on every vertex star, a Z check on every triangle. On the side-3 torus this is 972 qubits with every X and Z check overlapping on an even number of edges, and the rank computation over $\mathbb{F}_2$ gives exactly four logical qubits, the first Betti number of the four-torus [1, 4]. An open patch has zero. The prediction was four at every side, and the measurement returned eight at side 4. The explanation is the parity fact of Section 2: the even-sided lattice is two four-tori, each with its own four. The code counted components before anyone had looked for them. The mesh, then, is a substrate on which stabiliser codes live natively, and the hyperbolic versions [13] are the next thing to build on a closed quotient.

The Ward identity and the Wilson lines. Section 4 gave the numbers. What the reproduction taught is that the zero-mode count of a lattice gauge operator on a torus is not one gauge mode per site less one, but that plus the topology, $b_1 = 3$, which the naive prediction missed and the measurement supplied. A second experiment that tested only that the zero-mode count was positive now tests it against sites $+ 2$ exactly.

The Ehrenfest theorem, with a packet that is a particle. The walk's dispersion is $\cos E = \cos m \cos k$ with group velocity $v = \cos m \sin k / \sin E$, and the positive band is the eigenvector of the one-step unitary with eigenvalue e^{-iE}. A Gaussian packet built from that band alone, at momentum 0.6 and mass 0.5 on 400 sites, is 99.9999% positive-energy. The Ehrenfest prediction is computed from the packet's initial momentum distribution, not read off the run: the mean momentum shifts by exactly $-F$ per step under a linear potential, and the centroid advances by the distribution-averaged group velocity. Two facts had to be right for the prediction to hold. A band packet sits at the Berry connection of its band, -0.142 cells here, measured and predicted alike, and that offset moves as the force drifts the momentum. And the momentum kick lands between the coin and the shift, so each step sees the midpoint momentum $k - F(t + \frac{1}{2})$: with the start-of-step momentum the residual is 0.19 cells and grows linearly with the force, with the midpoint it is 0.0024. Free, the centroid matches to 9×10^{-14} cells over 57 cells of travel. Under a force of 0.005 per step it matches to 0.0024 cells over 45, with the momentum exact to 4×10^{-16} and a Zener leak of 2.4×10^{-5} into the other band. The controls are the two ways to break the theorem: a force of 0.8 leaves 26% of the weight in the band and the centroid 14 cells off, and the equal-chirality Gaussian "at rest", which the first attempt used and which followed no classical path, is 21% negative-energy and lags the classical trajectory by 19 cells. That seed was the reason the first probe failed, and it is kept as the control that explains why [14].

These are the shapes a good L2 has: an exact number, a control that breaks it, a prediction that was wrong once and corrected by the measurement, and the model named in the title.

Corrections to the Earlier Papers

The earlier paper [8] listed among its findings that quantum mechanics is derived, that spin-statistics holds on the same footing, and that holography is derived, and it described the walk results as measured on the coin's own Dirac walk. The audit corrects each.

- **The walk is a model, not the rule's sector.** Every walk result is a reproduction of known quantum-walk physics on a hand-written unitary map (Section 5). The rule has no single-particle sector with an amplitude. The sentence "the coin's own Dirac walk" is withdrawn.

- **Born rule, Tsirelson, CHSH, GHZ, contextuality.** These are exact results about small Hilbert spaces, the D_4 coin's quaternion units supplying an anticommuting pair. They are L1 and are cited as such. The base rule does not produce them.

- **Spin-statistics.** The exchange-phase result was a consistency check between assumed formulas, L0, and the exclusion result is L1 group arithmetic on the coin. The double cover (the twenty-four directions form the binary tetrahedral group, a spinor returns only at 4π) stands as mathematics.

- **Holography.** The area law and the Page curve run on free-fermion chains and Hilbert-space models. The five holography results that computed graph geodesics and shell fractions of hyperbolic space are known geometry with no dynamics, L2. Four holography results that run a tone dynamics on the mesh remain L3.

- **Controls that were constants.** The proton-lifetime control was a typed `false`, now the computed bare-Standard-Model lifetime. The area-law volume control was typed, now computed from the infinite-temperature correlation matrix. Twenty-two others of the same kind are gone.

- **Two passes were parity artifacts.** An all-or-none ignition experiment reported a "global" broadcast of exactly 0.5, the reachable half of an even-sided mesh, with a sub-threshold seed staying under 0.05. On a connected mesh the broadcast reaches 1.0 and the seed one radius below the critical one already spreads to 0.26, so the claim fails. A recordability-ceiling experiment had a plateau flatness of 0.013 on the split mesh and 0.173 on every connected mesh tried. Both now run on an odd side with their thresholds untouched and report the fail.

- **Emergent counts.** The earlier paper's scoreboard counted 92 emergent results. The count is 45, with 39 marked for publication. Quantum and spin have none.

What stands from the earlier paper, at the grade stated: the geometry of the mesh and its algebra (L1), the Standard Model representation content and $\sin^2\theta_W = 3/8$ as algebra of the octonions and F_4 (L1 to L2), the lattice gauge results as reproductions (L2), the genesis, growth, curvature-arrow, light-cone, isotropy, memory and self results as emergent (L3), and the three-generation identification as the open conjecture it was already stated to be.

Where the Theory Stands

The base is well defined, exactly reversible, and produces a light cone with a measured speed converging on one, conserved charges, hyperbolic geometry with a flat cusp on which compact selves can live, a growing mesh with a genesis attractor that forgets its start and needs negative curvature, a cell complex that carries stabiliser codes, rotational isotropy restored to fourth order by the D_4 symmetry with no random structure, and a coarse $\mathbb{Z}_3$ amplitude with a dynamical clock phase. It does not produce interference, and no phase on its configurations or on its pairs can. The clock phase the single-tone sector delivers is the same for every defect.

The sentence the program can now write is this: a deterministic discrete base with relativity, gauge structure, geometry and codes, and a classical particle where the quantum should be. Everything quantum in the suite is a reproduction on a model that the rule has not yet produced, and is labelled so. The single construction everything quantum waits on is a coarse variable, a count or density over many cells or beats, whose induced dynamics under the permutation is many-to-one and unitary on the coarse space, with an interaction that shifts one defect's clock against another's and a probability meaning for $|A|^2$. The three exact negatives say where it cannot be, which is what makes the search finite.

The Open Problems

Each problem is stated with the constraints the audit proved, because the constraints are what make it a problem rather than a hope.

1. The amplitude. Find a coarse variable A of the mesh state whose induced dynamics under one beat is many-to-one and unitary on the coarse space, and under which two seeded configurations show a site-wise cross term that the same map on a scrambled-neighbour mesh does not. Proved constraints: A cannot be a phase on configurations (the lift is a permutation matrix) nor on two-tone pairs at a cell (again a permutation). The $\mathbb{Z}_3$ sum over the mesh is a valid A that cancels on the vacuum and carries the clock, but every defect shares the clock's phase. So A must either average over many beats (the three-beat cycle is the natural window, and a defect born in a different beat of the cycle would carry a different phase), or over many cells with an interaction that advances one defect's clock relative to another's, or be a correlation function over many tones rather than a sum. The test is already written: the cross term of two seeds under the rule, against the scrambled-mesh control. What is missing is the variable.

2. The Born weight. Even with an amplitude, $|A|^2$ needs a probability meaning. The measurement arc shows that the emergent holder settles every run to a definite branch, but the fraction settling to one side is a step in the bias, not the proportional or squared weight. The weight and the collapse are separate mechanisms in this model, and the weight has no base-level origin yet.

3. Three generations with distinct masses. The exceptional Jordan algebra forces a rank-three frame with an exact S_3 family symmetry, so the count and the symmetry are geometric. The deterministic S_3-symmetric dynamics cannot break an exact symmetry spontaneously, and the only thing that splits the masses is an external asymmetric vacuum whose ratio is unconstrained. The hierarchy is a free input. The negative is exact and the conjecture stays open.

4. Spin-2 propagation. The base supports a stable, reversible, nonlinear propagating scalar field in the curved bulk. It does not yet support the graviton on the emergent cusp metric, which the gravity results locate as infrared-emergent. The reversible-gravity frontier is the radiative half, unsolved.

5. The interacting massless mode. Deterministic reversible rules give ballistic momentum, and the interacting massless mode needs a second conserved current, and the stochastic rule that was tried conserves only the $U(1)$ charge with no spontaneous order. The momentum rule was the answer to that negative, and its interacting sector is unmeasured.

6. Selection. The single outcome is deterministic, but selection among symmetric alternatives is not provided by the reversible rule (symmetric drains hold the pointer at zero). The amplifier is the arrow, measured as a positive Lyapunov exponent only with the arrow on, and the holder is the self. The tie from the amplifier to a specific branch weight is open.

7. Why this mesh. The substrate survey compares $\{3, 4, 3, 4\}$ with its rivals on a fixed battery, and the a priori argument for it runs through the pinch to dimension eight and the octonions. A forced derivation exists on paper [9]. Whether the dynamics selects it, rather than the algebra, is unmeasured.

8. The fills. Fourteen emergent results and 184 experiments in all seed with a hashed or pseudo-random fill, which the methodology admits only as an ensemble claim. Replacing every fill with a structured pattern, and showing the verdict survives, is unfinished.

9. Knife edges. The adaptive-fill self-emergence result, the anesthetic-collapse result, and the two size-floor results hold on one side of a threshold at one size. Each is recorded and none was tuned. Whether they are physics or artifacts of a thirteen-hundred-cell torus needs meshes an order of magnitude larger.

10. The label check's blind side. The substrate check flags an experiment that claims the mesh with no evidence. It does not flag an experiment that runs the rule on a square lattice and claims no substrate. A reverse pass is not written.

Directions

Time coarse-graining over the vacuum cycle. The vacuum clock has period three and every defect so far is born in the same beat of it. Seed defects at beats 0, 1 and 2 of the cycle and measure the $\mathbb{Z}_3$ amplitude's relative phase between them. If the phase difference is the clock difference, two defects of different birth beat would interfere in the coarse sum, and the next question is whether any interaction of the rule shifts a defect's birth beat. That is a two-experiment programme on code that exists.

Density waves of the momentum rule. The momentum rule's lone tone is ballistic, but its many-tone sector is a lattice gas with a conserved D_4 momentum, and a lattice gas has sound. The density wave of a lattice gas is a coarse variable that is many-to-one by construction. Measure its dispersion, its superposition of two waves, and whether the superposition carries a cross term the scrambled mesh lacks. The fluids arena already measures a sound speed near $1/\sqrt{2}$ on this rule, so the wave exists.

Correlation functions over many tones. A two-point function $\langle t_x t_y \rangle$ over the mesh is a coarse variable whose dynamics under a permutation is many-to-one. The Koopman lift has a spectrum on such observables even when it is a permutation on configurations, and the relevant question is whether any eigenfunction of the lift on the two-point sector carries a phase that depends on the seed's position. The lift on configurations is a permutation matrix and has only roots of unity for eigenvalues, but its restriction to the sector of two-point observables is not a permutation, and that is the space in which to look [3, 6].

A closed hyperbolic quotient. Every mesh in the suite is either a periodic flat D_4 lattice or a ball cut from the infinite hyperbolic honeycomb with a frontier. A compact quotient of $\{3, 4, 3, 4\}$ or $\{5, 3, 4\}$ by a torsion-free subgroup would give a finite exact mesh with no boundary and no frontier, on which the toric code, the Ward identity and the genesis attractor could be measured without a cut. Margenstern's machinery for cellular automata in hyperbolic spaces, the Fibonacci addressing of the pentagrid and the heptagrid and the navigation it supports [11, 12], is the route to building one, and it also reframes the program's addressing results as the four-dimensional case of an existing theory.

Qutrit codes on the D_4 complex. The tone is ternary. The toric code above used qubits on edges because the check matrices were built over $\mathbb{F}_2$. The same complex over $\mathbb{F}_3$ gives a $\mathbb{Z}_3$ toric code whose stabilisers are the tone's own group, and whose anyons are the ternary anyons the spin arena already counts. If the rule's conserved charge is a stabiliser check, which the conservation-as-stabiliser experiment shows for single-site errors, the qutrit code is the natural home for it.

Hyperbolic codes. On a closed quotient the same construction gives a hyperbolic surface code [13] with a constant encoding rate, and the mesh's exponential shell growth is what gives such codes their rate. Measuring the rate and distance of the code the honeycomb carries would be the first result that uses the curvature for something quantum rather than something geometric.

The rest of the cheap quantum rows. The harmonic oscillator spectrum on the tight-binding well, the adiabatic theorem and the Berry phase on the split-step walk, Goldstone's theorem on a lattice with a broken continuous symmetry, and coherent states on the walk are each a day's work on the existing library, each L2, each with prior art. They fill the coverage map and none of them touches the open problem.

Larger meshes for the knife edges. The selves and self-emergence results that flip at a size boundary run on tori of 625 to 2401 cells. The reflection engine reaches 4516 cells of $\{5, 3, 4\}$ at 90000 chambers in under a second, and the flat layer reaches fifteen million. Rerunning the four knife edges at ten times their size would settle whether they are physics.

Two experimental signatures. The isotropy order gives a q^6 anisotropy on D_4 against q^2 on a cubic lattice, unobservable at the gamma-ray bound but a definite difference in principle. The Bell-deviation result gives a power-law decline of the Bell violation beyond a crossover distance while quantum mechanics stays flat. Both are model predictions at L2 until the amplitude problem is solved, and both are stated so that they could be wrong.

Reproducibility

Every number in this paper is printed by an experiment in the program's open source repository [10], on the branch that carries this audit. Each experiment declares its identifier, its depth, its substrate label, its claim, its metrics and its computed control, and the runner reports a status of pass, fail, partial or open for each. The appendix indexes the experiments cited here. The suite's own checks (the depth labels, the substrate labels, the typed constants, the coverage map, the scale perturbation) run with the suite, and the type checker runs before it. The full run before the last edits of the audit reported 804 passes, 3 fails, 9 partials and 15 open results over 831 registrations with no crashes, the three fails being the bare causal-set action's known non-manifold dominance, a causal-set weighting result, and the self-emergence knife edge. The edits after that run are covered by before-and-after diffs of every affected experiment, and by the run recorded on the branch.

Claude Code, Anthropic's coding agent, was used heavily throughout this program. The earlier experiments and papers were written and iterated with the Claude Opus 4 and Opus 5 models, and this audit, its new experiments, its checks and this paper were written with Claude Fable 5. The author set the questions, the standards and the verdicts on what counts, and read the results. The code and the prose were produced with the models, and the reader should weigh the work knowing that.

Index of Experiments Cited

Identifiers as they appear in the suite's catalog. Depth is the grade after the audit.

identifier	depth	what it measures
E-FND-0080	L2	the charge rule has no amplitude: vacuum period three, a defect of two slots, exact union of two defects, against the walk's spread and cross term
E-FND-0081	L2	the momentum rule's lone tone is a ballistic classical particle, pairs are exact unions, head-on opposite signs pass through
E-FND-0082	L1	the lift of the rule to superpositions of configurations is a permutation matrix, 49 branches to 49 images, cross term zero
E-FND-0083	L1	the induced map on the 1104 two-tone pair states at a cell is a permutation
E-FND-0084	L2	the $\mathbb{Z}_3$ vacuum-clock amplitude: exact vacuum cancellation, defect phase locked to the clock, all defects in phase
E-FRC-0072	L2	the lattice Ward identity, sites + 2 zero modes on the three-torus, Proca control
E-QTM-0093	L2	the toric code on the D_4 complex, four logical qubits per connected component
E-QTM-0094	L2	the Ehrenfest theorem on the walk with a positive-band packet, Berry connection and midpoint momentum
E-CSM-0002	L3	the genesis attractor is canonical across starts on $\{5, 3, 4\}$
E-CSM-0008	L3	only negative curvature sustains the arrow
E-CSM-0013	L3	genesis on purely ternary integer $\{3, 4, 3, 4\}$ structure

E-CSM-0023	L3	every charge-zero start converges to the same living balance
E-CSM-0026	L3	a growing mesh self-creates with no arrow parameter
E-RLT-0018	L3	isotropy to fourth order from the D_4 moment identities, q^6 against q^2
E-RLT-0030	L3	the propagating mode's speed squared 0.959, 0.980, 0.985 at sides 12, 24, 36
E-GRV-0031	L3	a stable, reversible, nonlinear propagating field in the curved bulk
E-MMR-0011, 0013	L3	associative search cost and communication radius, logarithmic on the bulk
E-MMR-0007	L3	attractor recall on the dissipative layer, not on the bare rule
E-SLF-0056	L3	selves compact and persistent on the flat horosphere, built at fifteen million cells
E-SLF-0061, 0079, 0112	L3	tone integration reads the dynamics, the self of selves, the self-emergence knife edge
E-MMR-0002, E-SLF-0120, E-SLF-0161	L3, L3, L2	the size floors of capacity-versus-curvature and selves-at-scale, and the anesthetic-collapse knife edge
E-SLF-0166	L3	global-workspace ignition, a parity artifact, now a fail on a connected mesh
E-FND-0072	L2	the recordability ceiling, a parity artifact, now a fail on a connected mesh

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